

# YUJIA ZENG

✉ yujiazng@gmail.com · 🌐 zyjOrz · ★https://www.yujiazeng.com/ ·

## 🎓 EDUCATION

**University of Science and Technology of China, School of the Gifted Young**, Anhui 2023-Present

*Undergraduate Student* in Artificial Intelligence, Expected to graduate in June 2027

Academic Average: 91.16 / 100 (Top 5%)

## 🏛️ ACADEMIC ACHIEVEMENTS

- National Scholarship ( ¥ 10000, top 1% university-wide; highest national undergraduate honor)
- Yang Ya Alumni Fund Scholarship ( ¥ 5000, top 5 female students in School of the Gifted Young)
- AI Elite Class Scholarship ( ¥ 5800, awarded to the top 3 students in the Artificial Intelligence major)
- ICPC HongKong Regional Contest Silver Medalist (*International Collegiate Programming Contest*)

## 📄 PUBLICATIONS

1. **Yujia Zeng\***, Chensheng Peng\*, Yuxin Chen\*, Alex Shao, Nathan Jew, and Masayoshi Tomizuka.  
**RoboSTAR: Next-Scale Autoregressive Sign Language Motion Translation for Humanoid Robots.**  
Accepted as an **Oral Presentation** at the *IROS 2026 Workshop on Nonverbal Cues for Human-Robot Co-operative Intelligence*; under review at *ICLR 2027*. Equal contribution.
2. Bowen Xue, Brandon Y. Feng, Chenguo Lin, Yuchen Lin, **Yujia Zeng**, Lvmin Zhang, Maneesh Agrawala, Honglei Yan, and Panwang Pan.  
**Ring Forcing: Towards Precise Long-Term Memory for Autoregressive Video Diffusion.** *European Conference on Computer Vision (ECCV)*, 2026.
3. **Yujia Zeng.**  
**ReconNet: Generative Recommendation with Control-Guided Diffusion Models.** *International Conference on Intelligent Computing (ICIC)*, 2026. **Oral Presentation.**
4. Ying Yang, Zhengyao Lv, **Yujia Zeng**, Tianlin Pan, Haofan Wang, Yueming Lyu, Binxin Yang, Hubery Yin, Chen Li, Jing Lyu, Ziwei Liu, and Chenyang Si.  
**StableWorld: Towards Stable and Consistent Long Interactive Video Generation.** *arXiv preprint*, 2026.
5. Chensheng Peng, Chenpeng Su, Chengwei Zhang, Jiuming Liu, **Yujia Zeng**, Yixiao Wang, Chenfeng Xu, Yichen Xie, Thomas Monninger, Zhipeng Mo, Sihao Ding, Wenzhao Zheng, Kurt Keutzer, Masayoshi Tomizuka, and Wei Zhan.  
**GenRec4D: Hybrid 4D Gaussian Reconstruction with Feedforward and Generative Priors.** *Under review at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*.

## 👤 EXPERIENCE

**UC Berkeley**   MSC Lab   Supervisor: Prof. Masayoshi Tomizuka   Mar. 2026 – Mar. 2027  
*Generative Models, Robotics, Dexterous Hand*   J-1 Research Intern

- Proposed **RoboSTAR**, a next-scale autoregressive framework for speech/text-conditioned sign language motion translation for humanoid robots, achieving state-of-the-art performance.
- Contributed to **GenRec4D**, a hybrid 4D Gaussian reconstruction framework that combines feedforward reconstruction with generative priors for high-quality dynamic scene reconstruction.

**Harvard University**   Embodied Minds Lab   Supervisor: Prof. Yilun Du   Sep. 2026 – Present  
*Dexterous Manipulation, Human-Object Interaction*   Research Intern

- Conducting research on generative human-object interaction and dexterous manipulation, in collaboration with the MSC Lab at UC Berkeley.

Supervisor: Prof. Ziwei Liu, Prof. Chenyang Si  
World Models, Video Generation Research Project

- Presented **StableWorld**, investigated long-horizon interactive video generation, focusing on error accumulation and temporal drift in autoregressive world models.
- Developed a model-agnostic dynamic frame eviction mechanism that filters degraded historical frames while retaining geometrically consistent context for more stable long-term generation.

**LDS** Supervisor: Prof. Xiangnan He, Prof. Xiang Wang

Oct. 2025

Diffusion, Recommendation Systems Individual Research Project

- Designed **ReconNet**, reformulating sequential recommendation as a control-guided diffusion generation problem, using multi-domain user preferences as controllable signals.
- Designed residual-control and cross-norm mechanisms to guide personalized item-embedding generation.

**Baidu**

Mar. 2025 – Jul. 2025

AIGC, Reinforcement Learning Algorithm Intern

- Applied DPO-based preference optimization to improve image-generation models for business-facing applications.
- Contributed to model iteration and image-generation engineering workflows, bridging generative modeling with practical product requirements.

**HiDream**

Sep. 2025 – Jan. 2026

Generative AI, Image Editing Algorithm Intern

- Explored instruction-guided image editing and controllable generation for fine-grained visual manipulation and refinement.
- Prototyped generative-AI workflows for design-oriented products, including automated website design and visual content creation.

⚙️ IT SKILLS

- Programming: Python, C/C++, PyTorch
- Robotics & Tools: ROS/ROS2, MuJoCo, Isaac Lab, Linux, Git, Docker

📖 CORE COURSES

**Artificial Intelligence & Computer Science:**

|                                         |    |                                    |    |
|-----------------------------------------|----|------------------------------------|----|
| Introduction to Artificial Intelligence | 97 | AI & Machine Learning Fundamentals | 97 |
| Artificial Intelligence Practice        | 95 | C Programming Language             | 97 |
| Scientific Programming Practice         | A+ | Academic Writing                   | A+ |

**Mathematics & Physics:**

|                 |    |                 |     |
|-----------------|----|-----------------|-----|
| Linear Algebra  | 96 | Thermal Physics | 100 |
| Quantum Physics | 95 | Mechanics       | 95  |

📌 OTHERS

- CET-4: 623, CET-6: 610
- President of the Student Union, School of the Gifted Young
- the Outstanding Physical Fitness Award (top 1% university-wide)